RV College of Engineering®
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Academic year 2025-2026 (Odd Sem)
DEPARTMENT OF

Computer Science & Engineering

Date	21st Jan 2025	Maximum Marks	10+50
Course Code	CS354TA	Duration	120 Min
Sem-V	ReTest	Staff: MRA|PK	

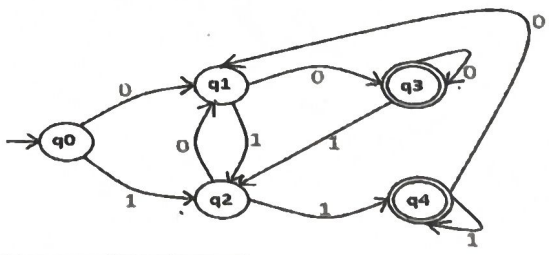
THEORY OF COMPUTATION
(Common to CSE & ISE)

PART-A

		Marks	BT	CO
1.1	Define left recursion. Eliminate left recursion from the following grammar A → ABd / Aa / a B → Be / b	2	L2	CO4
1.2	For the following CFG, identify the null productions and the unit productions. S → ABCBCDA, A → CD, B → Cb, C → a | Є, D → bD | Є	2	L2	CO3
1.3	Identify the language accepted by the following CFG. S → Ab | aB | Є A → Sa B → bS	1	L3	CO2
1.4	If L1 is CFL and L2 is regular, then L1∩L2 is _____	1	L3	CO2
1.5	Construct DPDA equivalent to the DFA whose transition table is as below: Here sate A is the start state and state C is the final state.	2	L1	CO1
1.6	Define DPDA.	2	L1	CO1

ε	0	1
A	B	A
B	B	C
C	B	A

PART-B

2.a	Construct PDA that accepts the language $L = \{ w\, c\, w^R \mid w \in \{a, b\}\}^*$. Give the graphical representation for PDA obtained. Show the moves made by the PDA for the string **abbcbba**..	7	L3	CO4
2.b	Construct a finite automaton that accepts the language generated by the right linear grammar S -> aA | Ɛ, A -> aA | bB | Ɛ, B -> bB |Ɛ	3	L2	CO1
3.a	Find a left-linear grammar for the language accepted by the finite automata below  	6	L3	CO3
3.b	Write a note on the languages accepted by PDA.	4	L1	CO2
4.a	Define CNF. Transform the grammar with productions **S → AB | ε** **A → aA | a | ε** **B → bB | b | C** **C → cC | ε** **D → d** into Chomsky normal form.	6	L3	CO3
4.b	Define ambiguous grammar. Show that the following grammar is ambiguous and Construct an unambiguous grammar equivalent to this grammar.	4	L2	CO4

	$S \rightarrow aS \mid bA \mid \varepsilon$ $A \rightarrow aA \mid bS$			
5	Give an algorithm to convert PDA to CFG. Convert the PDA with transitions $\delta(q0,a,Z)=(q0,AZ)$, $\delta(q0,b,Z)=(q0,BZ)$, $\delta(q0,c,A)=(q0,\varepsilon)$, $\delta(q0,c,B)=(q0,\varepsilon)$, $\delta(q0,a,A)=(q0,AA)$, $\delta(q0,b,B)=(q0,BB)$, $\delta(q0,\varepsilon,Z)=(q1,\varepsilon)$ to a CFG. Describe the language accepted by the CFG.	10	L4	CO3
6a	Let L1= {ww | w ∈ {a,b}* } and L2={ ~~wRw~~ wwR | w ∈ {a,b}* }. Find L3=L1∩L2 and L4=L1UL2. Show that L1 is not CFL and L2 is CFL.	6	L2	CO2
6 b	Show that the language L = { $a^i b^j c^k$ | i<j and j<k} is not context free.	4	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	13	24	13	7	21	22	10	-	-

Course Outcomes:

1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for better results.
4. Apply automata skills in situations that describe computation effectively and efficiently.

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